

Linear Dependence and Independence

Linear Dependence and Linear Independence

Definition 10.2.4. A set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ of vectors in a vector space V is called **linearly independent** if the following vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$$

has the only trivial solution $c_1 = c_2 = \dots = c_n = 0$. If the vector equation has any nontrivial solution, then S is called **linearly dependent**.

In other words, S is called **linearly independent** if none of the vectors in S is a linear combination of other vectors in S . Otherwise, S is called **linearly dependent**.

Example 10.2.6. Show that $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is linearly independent.

Solution. Consider the following equation

$$\begin{aligned} c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 &= \mathbf{0} \\ c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

Therefore, $c_1 = c_2 = c_3 = 0$. Hence S is linearly independent.

Example 10.2.7. Show that $S = \left\{ \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{\mathbf{v}_1}, \underbrace{\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}}_{\mathbf{v}_2}, \underbrace{\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{v}_3} \right\}$ is linearly independent.

Solution. Consider the following equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$$

The augmented matrix takes the form

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{array} \right] \quad [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \mid \mathbf{v}]$$

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By reducing it to the reduced row echelon form, we obtain

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{array} \right] &\xrightarrow{\substack{-2R_1+R_2 \rightarrow R_2 \\ -3R_1+R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 2 & 7 & 0 \end{array} \right] &\xrightarrow{-2R_2+R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & -1 & 0 \end{array} \right] \\ &\xrightarrow{-R_3 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] &\xrightarrow{\substack{-4R_3+R_2 \rightarrow R_2 \\ 2R_3+R_1 \rightarrow R_1}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \end{aligned}$$

This shows the system has the only trivial solution $c_1 = c_2 = c_3 = 0$. Therefore, S is linearly independent.

Example 10.2.8. Check if $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$ is linearly independent.

Solution. To solve the following vector equation

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{0},$$

we apply the same procedure as in Example 10.2.7 to the following matrix. By reducing it to the reduced row echelon form, we obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{array} \right] \xrightarrow{\substack{-2R_1+R_2 \rightarrow R_2 \\ -3R_1+R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 2 & 4 & 0 \end{array} \right] \xrightarrow{-2R_2+R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

This shows that c_3 is free variable, hence the system has a nontrivial solution.

$$c_1 - c_3 = 0$$

$$c_2 + 2c_3 = 0$$

For example, if $c_3 = 1$, then $c_1 = 1$ and $c_2 = -2$. That is

$$\mathbf{v}_1 + (-2)\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Hence S is linearly dependent.

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Remark If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{0}\}$, then S is linearly dependent. In other words, if a set contains the zero vector, it must be linearly dependent. Because the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k + c_{k+1}\mathbf{0} = \mathbf{0}.$$

has nontrivial solution, namely

$$0\mathbf{v}_1 + 0\mathbf{v}_2 + \cdots + 0\mathbf{v}_k + (1)\mathbf{0} = \mathbf{0}.$$

Theorem 10.2.2. A set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_i, \dots, \mathbf{v}_n\}$ is linearly dependent if and only if at least one of the vectors in S is a linear combination of the other vectors in S .

Proof. See Appendix.

Corollary A set $S = \{\mathbf{v}_1, \mathbf{v}_2\}$ of two nonzero vectors in V is linearly dependent if and only if $\mathbf{v}_2 = c\mathbf{v}_1$ for some scalar c .